# SQL FUNCTIONS

**1: String Functions - Clean up employee names for reporting**

```
SELECT Employee_name, UPPER(Employee_name) AS UpperName FROM Employees;
```

| Employee_name | UpperName |
|---|---|
| Arun Kumar | ARUN KUMAR |
| Priya Sharma | PRIYA SHARMA |
| Rahul Singh | RAHUL SINGH |
| Meena Reddy | MEENA REDDY |
| Vikram Das | VIKRAM DAS |

```
SELECT Employee_name, LOWER(Employee_name) AS LowerName FROM Employees;
```

| Employee_name | LowerName |
|---|---|
| Arun Kumar | arun kumar |
| Priya Sharma | priya sharma |
| Rahul Singh | rahul singh |
| Meena Reddy | meena reddy |
| Vikram Das | vikram das |

```
SELECT Employee_name, SUBSTRING(Employee_name,1,4) AS SubName FROM Employees;
```

| Employee_name | SubName |
|---|---|
| Arun Kumar | Arun |
| Priya Sharma | Priy |
| Rahul Singh | Rahu |
| Meena Reddy | Meen |
| Vikram Das | Vikr |

```
SELECT Employee_name,Gender,CONCAT(Employee_name, ' - ', Gender) AS EmpName_Gender FROM Employees;
```

| Employee_name | Gender | EmpName_Gender |
|---|---|---|
| Arun Kumar | M | Arun Kumar - M |
| Priya Sharma | F | Priya Sharma - F |
| Rahul Singh | M | Rahul Singh - M |
| Meena Reddy | F | Meena Reddy - F |
| Vikram Das | M | Vikram Das - M |

**2: Date Functions - Calculate employee tenure in years**

```
SELECT Employee_name,Hire_date,YEAR(NOW()) - YEAR(hire_date) AS TenureYears FROM Employees;
```

| Employee_name | Hire_date | TenureYears |
|---|---|---|
| Arun Kumar | 2012-05-14 | 14 |
| Priya Sharma | 2015-07-22 | 11 |
| Rahul Singh | 2010-03-10 | 16 |
| Meena Reddy | 2018-11-05 | 8 |
| Vikram Das | 2013-09-18 | 13 |

```
SELECT Employee_name, Hire_date,
ROUND(DATEDIFF(NOW(), hire_date) / 365, 1) AS TenureYears
FROM Employees;
```

| Employee_name | Hire_date | TenureYears |
|---|---|---|
| Arun Kumar | 2012-05-14 | 13.9 |
| Priya Sharma | 2015-07-22 | 10.7 |
| Rahul Singh | 2010-03-10 | 16.1 |
| Meena Reddy | 2018-11-05 | 7.4 |
| Vikram Das | 2013-09-18 | 12.5 |

**3: User-defined Function - Create a reusable function to return full name of a student**

**Function Created:**

```
DELIMITER //
CREATE FUNCTION GetNameWithGender (
    EmpName VARCHAR(50),
    Gender VARCHAR(10)
)
RETURNS VARCHAR(100)
DETERMINISTIC
BEGIN
    RETURN CONCAT(EmpName, ' - ', Gender);
END //
DELIMITER ;
```

**Calling Function:**

```
SELECT Employee_id,Employee_name,Gender,
       GetNameWithGender(employee_name, gender) AS FullName
FROM Employees;
```

| Employee_id | Employee_name | Gender | FullName |
|---|---|---|---|
| 1 | Arun Kumar | M | Arun Kumar - M |
| 2 | Priya Sharma | F | Priya Sharma - F |
| 3 | Rahul Singh | M | Rahul Singh - M |
| 4 | Meena Reddy | F | Meena Reddy - F |
| 5 | Vikram Das | M | Vikram Das - M |